

INCIDENT & VULNERABILITY LAB REPORT

OFFLINE PDF HASH CRACKING ANALYSIS

SUBJECT

Offline PDF Hash Cracking Analysis

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Classification	TRAINING

1. Executive Objectives

- Demonstrate the process of recovering passwords from protected PDF files by extracting their hashes and using dictionary attacks.
- Compare the methodology of using a local installation of John the Ripper (JTR) with the web-based Networkwalks tools.

2. Environment & Tools

The following operating system and specialized tools were used across both modules to conduct hash extraction and dictionary attacks.

Category	Details
Operating System	Windows
Hash Extraction Tools	Online Hash Extractor (onlinehashcrack.com) Networkwalks Hash Calculator
Password Cracking Tools	John the Ripper (JTR) / Johnny GUI Networkwalks Password Cracker

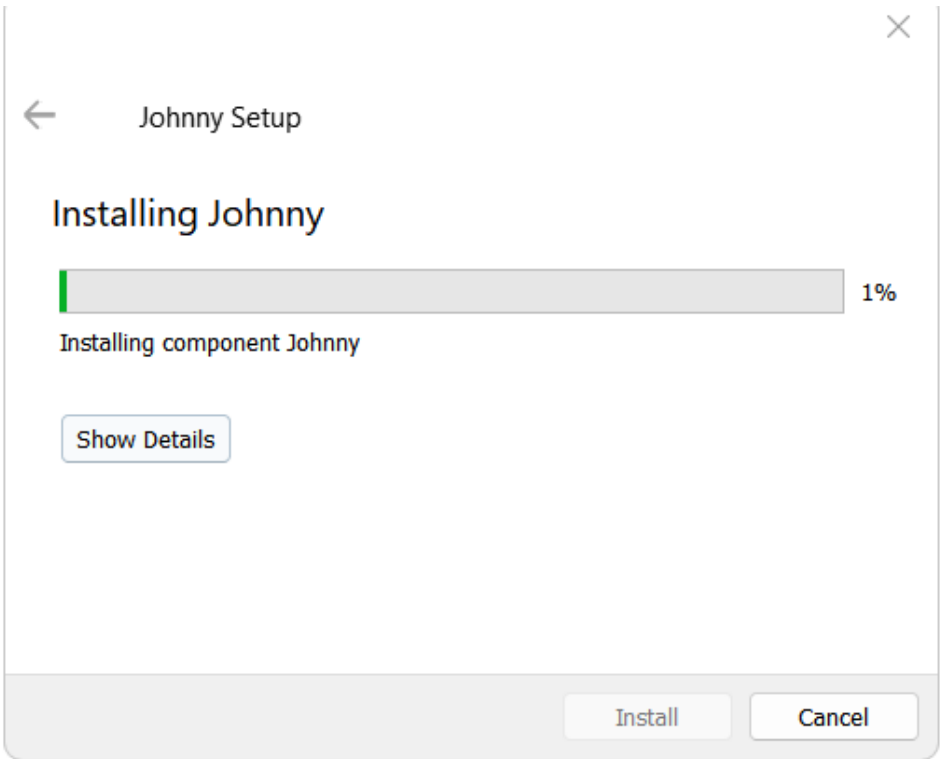


Figure 1. Installation of Johnny.

3. Methodology

3.1 Module 1 - John the Ripper (JTR)

- Hash Extraction: The online tool at onlinehashcrack.com was used to extract the \$pdf\$ hashes from the locked files.
- Preparation: The extracted hash values were saved into standard .txt files (e.g., hash1.txt.txt, hash2.txt, hash3.txt) to prepare them for local processing.
- Cracking Process: The text files were imported into the Johnny GUI, which serves as the graphical interface for JTR, and a dictionary attack was executed to recover the plaintext passwords.

3.2 Module 2 - Networkwalks Tools

- Hash Extraction: The target PDFs were uploaded directly to the Networkwalks Hash Calculator to parse the files and extract their hashes within the web browser.
- Cracking Process: The extracted hashes were pasted into the Networkwalks Password Cracker, and its built-in wordlist attack was executed to recover the passwords.

4. Results & Evidence Log

The dictionary attacks successfully recovered the passwords for all three target PDF files and enabled the associated flags to be captured. The hash values shown below are intentionally truncated, consistent with the original evidence log.

Target	Recovered Password	Captured Flag	Methods Used
1	good-luck	nw{cybersecurity_flag_captured_2608}	JTR / Johnny; Networkwalks Password Cracker
2	password1	nw{networkwalks_persistence_jtr_270521}	JTR / Johnny; Networkwalks Password Cracker
3	1qaz2wsx	nw{networkwalks_flag_260821_1}	JTR / Johnny; Networkwalks Password Cracker

Target 1

Target	Target 1
Extracted Hash	\$pdf\$4*4*128*-1028*1*16*ca7f72f...
Cracked Password	good-luck
Captured Flag	nw{cybersecurity_flag_captured_2608}

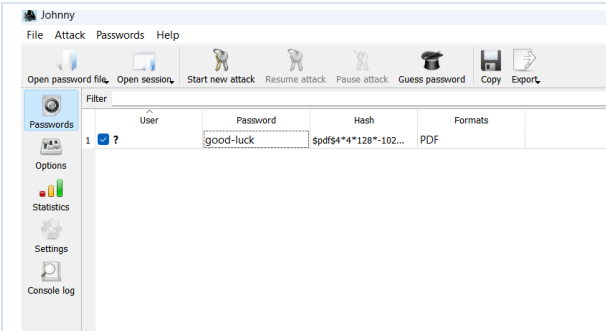


Figure 2. Johnny result for Target 1.

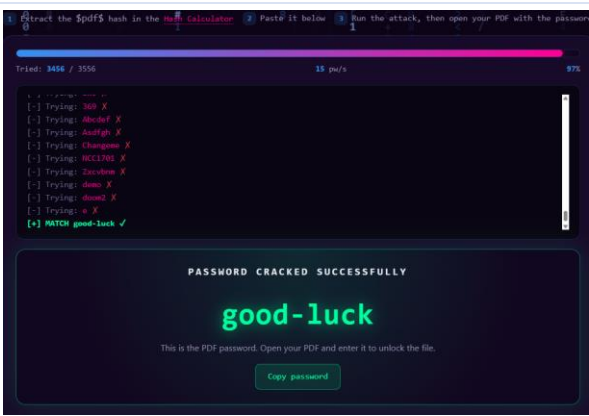


Figure 3. Networkwalks password-cracker result for Target 1.

Captured Flag: nw{cybersecurity_flag_captured_2608}



Figure 4. Captured flag evidence for Target 1.

Target 2

Target	Target 2
Extracted Hash	\$pdf\$4*4*128*-1028*1*16*0853f2c...
Cracked Password	password1
Captured Flag	nw{networkwalks_persistence_jtr_270521}

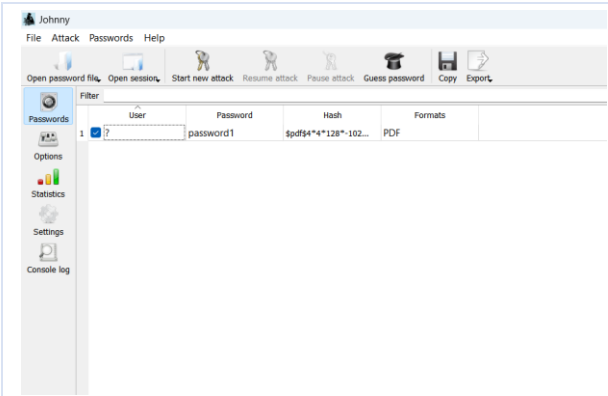


Figure 5. Johnny result for Target 2.

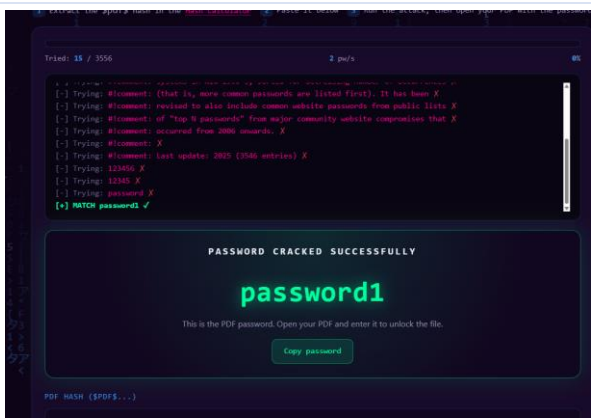


Figure 6. Networkwalks password-cracker result for Target 2.

Captured Flag: nw{networkwalks_persistence_jtr_270521}

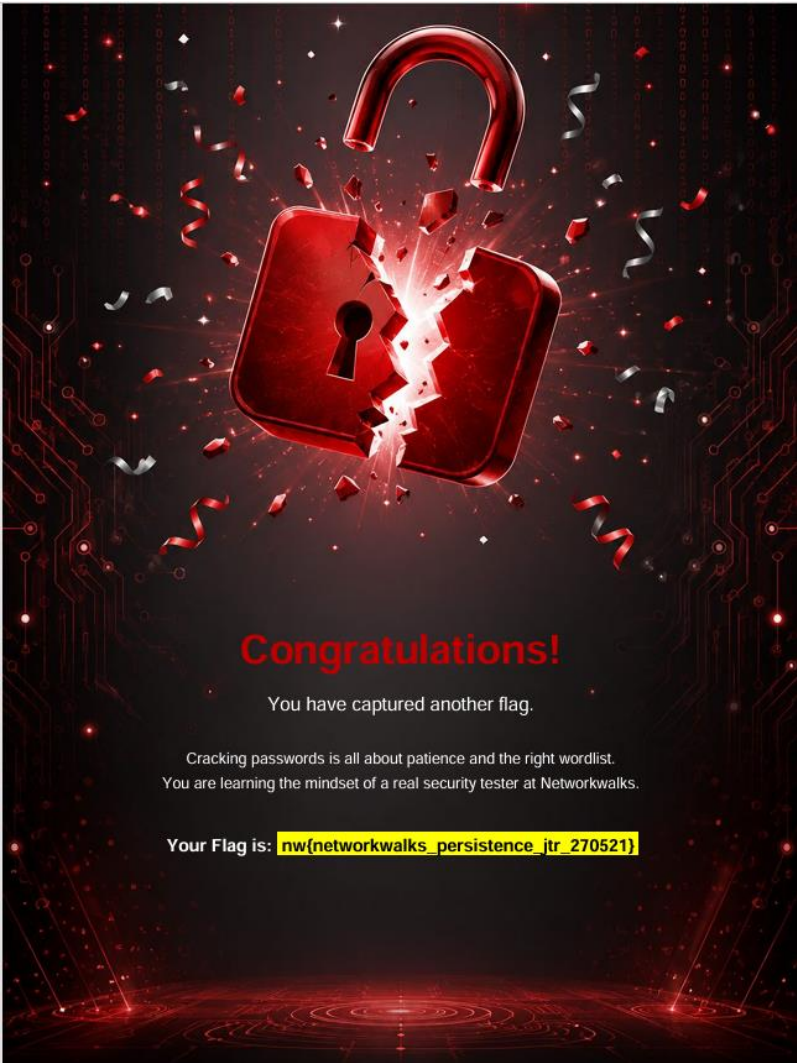


Figure 7. Captured flag evidence for Target 2.

Target 3

Target	Target 3
Extracted Hash	\$pdf\$4*4*128*-1028*1*16*34eb542...
Cracked Password	1qaz2wsx
Captured Flag	nw{networkwalks_flag_260821_1}

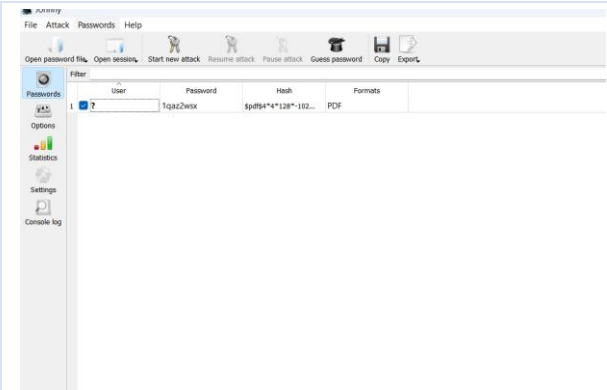


Figure 8. Johnny result for Target 3.

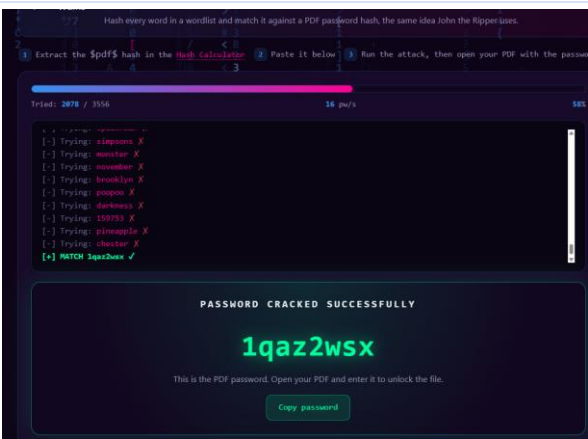


Figure 9. Networkwalks password-cracker result for Target 3.

Captured Flag: nw{networkwalks_flag_260821_1}

Congratulations!

You have captured your flag.

Let's continue this amazing journey of Cybersecurity with incredible team of experts at Networkwalks.

Your Flag is: nw{networkwalks_flag_260821_1}

You can also share this achievement on your LinkedIn & tag Networkwalks & your instructor to share your achievement.



Figure 10. Captured flag evidence for Target 3.

5. Mitigation & Remediation Strategies

To reduce the likelihood of successful offline dictionary attacks, the following technical controls and policies should be implemented:

- **Enforce Strong Passphrases:** Dictionary attacks rely heavily on common words and simple alphanumeric sequences. Using long passphrases (15+ characters) with high entropy makes dictionary and brute-force attacks substantially more difficult.
- **Avoid Predictable Patterns:** Users should avoid standard keyboard walks such as 1qaz2wsx or appending numbers to common words such as password1, as modern wordlists commonly test these permutations.
- **Strong Encryption Standards:** Ensure that files are secured using robust algorithms (e.g., AES-256) rather than legacy encryption methods, increasing the time required for cracking attempts.
- **Certificate-Based Security:** For highly sensitive documents, use certificate-based encryption rather than password-based encryption to reduce reliance on human password selection.

6. Conclusion

The exercises demonstrated that predictable passwords such as password1 and 1qaz2wsx can be recovered using standard wordlists and readily available tools. The results reinforce the importance of strong password selection, avoidance of predictable patterns, and appropriate document-encryption controls when protecting files against offline hash-cracking attempts.